

AIoT Mangosteen Ripeness Detection

Welcome to the 3-Week AIoT Mini-Project! In this project, two groups will jointly build a dataset, but independently engineer and deploy fruit ripeness detection AI on an ESP32-S3.

🌟 The Objective

Build a complete AIoT edge-device prototype using an ESP32-S3 and a camera to correctly classify mangosteen ripeness into one of three categories: **Unripe, Ripe, or Overripe**.

⚙️ How It Should Be Done (The Project Twist)

This is not a standard isolated project. This project mirrors real-world ML engineering. It is divided into two distinct phases:

Phase 1: Collaborate on Data (Week 1)

- **What:** Both groups work **together in the same room**.
- **Action:** You will jointly buy fruit, calibrate cameras, take photos, and agree on labels.
- **Result:** Both groups start with the exact same 100% identical dataset.

Phase 2: Compete on Engineering (Weeks 2 & 3)

- **What:** Groups separate and work **completely independently**.
- **Action:** You will design your own CNN model, choose your own hyperparameters, write your own ESP32-S3 firmware, and size your own memory arena. Do not share code.
- **Result:** Different engineering choices will lead to different latencies, memory usage, and accuracies.

❓ The Ultimate Question

At the final joint presentation, both groups will put their benchmark cards side-by-side on the projector. The goal of this project is to answer this single question:

*"Two teams, same dataset, same target hardware, same benchmarking protocol — **why did our results diverge?**"*

The Big Question: *Two teams, same dataset, same hardware — why did our results diverge?*

Project Timeline briefly

Phase	Duration	Focus	Key Deliverable
Module 1	Week 1	Shared Dataset (Joint Work)	 Final labeled dataset (train/val/test)
Module 2	Week 2	Model & Deploy (Independent)	 Quantized TFLite model on hardware
Module 3	Week 3	Benchmark & Report (Independent)	 Benchmark Card & Final Presentation

Grading Breakdown

- **Report (60%):** Dataset docs, architecture rationale, hardware deployment, benchmark rigor, and honest failure analysis.
- **Presentation & Demo (40%):** Live on-device demo, presentation delivery, Colab reproducibility, and group comparison.

Quick Glossary (Important Terms)

- **Flash:** Storage space on the ESP32-S3 for your .tflite model file.
- **SRAM (Tensor Arena):** Working memory required during active inference.
- **int8 Quantization:** Converting float32 weights to 8-bit integers to fit edge hardware.
- **Inference Latency:** Milliseconds taken to classify one image.

Module 1: Week 1 - Dataset Foundation (Joint Work)

 **Important:** Both groups must work together this week to build a shared baseline dataset (Unripe, Ripe, Overripe). If this is done correctly, everything else becomes easier.

Roles

- **Data Lead:** Ensures photography protocol is strictly followed.
- **Model & Hardware Leads:** Assist with photography and cross-group labeling.

Weekly Task Checklist

- [] **Day 1-2:** Agree on the taxonomy and calibrate phone cameras (25 cm distance, no flash).
- [] **Day 3-5:** Purchase 50–60 mangosteens. Photograph each fruit 6 times (3 backgrounds × 2 lightings).

- [] **Day 6:** Cross-verify labels with the other group. Resolve any disagreements together.
- [] **Day 7:** Compile the final dataset (approx. 600 train/val images, 100 test images).

Assignment Submission for Week 1

- **Submit:** A link to the shared Google Drive folder containing the final dataset.
- **Submit:** The Signed Taxonomy Agreement & Photography Checklist.

Module 2: Week 2 - Model Development & Deployment (Independent)

 **Important:** From this point forward, groups work independently. Do not share code or model weights.

Roles

- **Model Lead:** Owns Colab, CNN architecture ($\leq 100,000$ parameters), and int8 quantization.
- **Hardware Lead:** Wires OV2640 and INA219 to ESP32-S3, integrates TFLite Micro.

Weekly Task Checklist

- [] **Colab:** Train a float32 CNN and achieve $\geq 88\%$ validation accuracy.
- [] **Colab:** Convert the model to an int8 .tflite file. Verify the accuracy drop is minimal ($< 2\%$).
- [] **Hardware:** Wire the ESP32-S3 and implement crop/downsample preprocessing to exactly match Colab.
- [] **Hardware:** Run the first successful on-device inference (even if accuracy is low).

Assignment Submission for Week 2

- **Submit:** A screenshot of your first successful on-device inference (UART output or LED/OLED).

Module 3: Week 3 - Benchmarking & Reporting (Independent)

 **Important:** This week is about measuring reality against expectations. Honest reporting of failures scores higher than fabricated success.

Roles

- **Hardware Lead:** Measures median/P95 latency and active/sleep energy.
- **All Members:** Write report sections and prepare presentation slides.

Weekly Task Checklist

- [] **Measure:** Record on-device test set accuracy and confusion matrix.
- [] **Measure:** Complete the Benchmark Card (Latency, Memory, Throughput, Energy).
- [] **Write:** Draft the "Failure Modes & Lessons Learned" report section (Bugs > 2 hours, abandoned architectures).
- [] **Present:** Finalize the 10-slide presentation and prep for the joint live demo.

Final Assignment Submissions

- **Submit 1:** Final Technical Report (PDF, 8-15 pages).
- **Submit 2:** Presentation Slides (PDF or PPT).
- **Submit 3:** Link to reproducible Google Colab Notebook.

Resources & Templates

Report Structure Outline (Fill-in-the-blanks)

Use this structure for your final PDF report:

1. **Abstract:** 150-word summary of project and findings.
2. **Intro & 3. Related Work:** 3–5 IEEE citations.
3. **Dataset:** Class distribution, sample images, agreement stats.
4. **Model & 6. Quantization:** Architecture table, training curves, accuracy comparison.
5. **Hardware & 8. Benchmarks:** Firmware pipeline, arena sizing, Benchmark Card table.
6. **Failure Modes (Crucial):** Bugs > 2 hours, abandoned architectures, hardware glitches.
7. **Group Comparison:** Side-by-side benchmark differences (done after the joint presentation).
8. **Conclusion:** 2–3 concrete extensions.

Presentation Slide Guide (10 Slides Total)

- **Slides 1–3:** Title, Motivation, Dataset Overview.
- **Slides 4–6:** Model Architecture, Quantization Results, Hardware Setup.
- **Slide 7:** Completed Benchmark Card.
- **Slide 8:** Live Demo & One Key Failure.
- **Slide 9:** Standardized Comparison Table (Compare with the other group).
- **Slide 10:** Conclusions & Future Work.

